PRECISION INDUSTRIES®

LIGHTING THE WORLD

PI Model	Hz/rpm	Voltage	Prime	Standby
PI 220C	50/1500	400/230	200 kVA 160 kWe	220 kVA 176 kWe
PI 250C	60/1800	380/220	225 kVA 180 kWe	250 kVA 200 kWe

Applicable Standard and References: ISO 8528-1; ISO 3046-1; BS 5514-1
The above given ratings are at 0.8 power factor (PF)

Engine Technical Data					
Engine Make & Model		Cummins QSB7-G5			
No. of Cylinders		6 In-line			
Cycle		4			
Aspiration		Turbocharged & Charged Air Cooled			
Cooling Method		Water			
Governing Type		Mechanical			
Governing Class		ISO 8528 Part 1 G2			
Compression Ratio		17.2 : 1			
Displacement (L)		6.69			
Bore/Stroke (mm)		107/124			
Battery Charger, Ampere		12 Volts DC,100 Amp			
Air System		**50Hz**	**60Hz**		
Air Filter Type		Dry Element (replaceable)			
Combustion Air Flow m³/min (cfm)	Standby	6.0 (212)	7.6 (269)		
	Prime	5.8 (205)	7.2 (256)		
Radiator Air Flow m³/min (cfm)		318 (11230)	379.2 (13391)		
Cooling System		**50Hz**	**60Hz**		
Total Coolant capacity (L)		26			
Water Pump Type		Dry Element (replaceable)			
Radiator Fan Load kW (hp)		6.9 (9.24)	12.7 (17.0)		
Heat Radiation to Room kW (BTU/min)	Standby	21 (1163)	24 (1342)		
	Prime	19 (1032)	21 (1154)		
Lubrication system					
Oil Filter Type		Spin on full flow filter, 25 micron			
Total Oil Capacity l (US gal)		18.9 (4.97)			
Oil Pan l (US gal)		15.1 (4.0)			
Oil Type		API CH4/CI4; SAE 15W-40			
Cooling Method		Water Cooled			
Fuel System		**50Hz**	**60Hz**		
Fuel Filter Type		Spin on fuel filters with water separator			
Recommended Fuel		Class A2 Diesel			
Fuel Consumption: l/hr (US gal/hr)		Prime	Standby	Prime	Standby
	110%	51 (13.4)	51 (13.4)	59 (15.5)	59 (15.5)
	100%	45 (11.9)	N.A	50 (13.3)	N.A
	75%	36 (9.5)	N.A	40 (10.6)	N.A
	50%	26 (6.9)	N.A	30 (7.8)	N.A
Exhaust System		**50Hz**	**60Hz**		
Silencer Size & Model		(2") Industrial			
Silencer Noise Reduction Level		11-15 dBA			
Max. Back Pressure kPa (in. Hg)		10.2 (3.0)			
Exhaust Gas Flow L/s (cfm)	Standby	597 (1265)	732 (1549)		
	Prime	569 (1205)	634 (1342)		
Exhaust Gas Temperature °C (°F)	Standby	561 (1041)	532 (988)		
	Prime	544 (1011)	487 (907)		

Alternator Technical Data		
Alternator Make & Model	Stamford UCI274H	
No. of Poles	4	
WINDING LEADS	12	
No. of Bearings	Single	
Insulation Class	Class H	
Winding Pitch	TWO THIRDS	
Ingress Protection Rating	IP-23	
Excitation System	SELF EXCITED	
AVR Model	AS440	
Alternator operating Data	**50Hz**	**60Hz**
Overspeed (RPM)	2250	
Voltage Regulation	± 1.5 %	
Wave Form NEMA = TIF	<50	
Wave Form IEC = THF	<2%	
Total Harmonic Content LL/LN	2%	
Radio Interference	Standard EN61000-6-2:2001	
Cooling Air Flow m³/sec(cfm)	30.84 (1090)	30.84 (1090)
Controller Technical Data		
Controller Make & Model	DSE 4520 or Eq.	
Operation mode	Auto Main Failure	
Display	Graphic back-lit LCD (128x64)pixels	
Ingress Protection Rating	Sealed front face(IP65)	
Binary Inputs/Outputs	4 / 6	
Analog inputs	4	
Measurement	Vac, A, Hz, kVA, kW, Vdc, KWh	
Event Log	Alarms log, Hours log	
Communication (Optional)	USB, RS232, RS485, GSM, ETHERNET	
Gen-set Enclosure Specification (optional)		
Enclosure Type	Acoustic and Weather Proof	
Anticorrosive Protection	Polyester Powder Coated Galvanized Sheet	
Ingress Protection rating	Weather Proof IP23	
Transportation	ISO standard lifting	
Noise level	72 dbA @ 7 meters	
Emergency	External Emergency Push Botton	
Canopy RAL color	RAL 2000	
Chassis RAL color	RAL 9011	
Dimensions	**Open Type**	**Enclosure**
Length (mm)	2500	3745
Width (mm)	1100	1165
Height (mm)	1752	2118
Weight (kg)	1950	3250
Fuel Tank Capacity (L)	550	550

Notes:

1-Prime power rating of generating set is a variable load and unlimited hours usage are applied on the generating set with an average load factor of 70% of the prime rating over each 24 hour period. Noting that a 10% overload is available for 1 hour in every 12 hours operation.
2-Standby power rating of the generating set is a variable load limited to an annual usage upto 500 hours is applied, with 300 hours of which may be continuous running. Noting that no overload is is permitted.
3- Warranty: one year or (1000) hours which comes first.
4- Referring to our company policy of continuous development, PI reserves the right to change specification without notice.

N.A. - Not Available
N/A - Not Applicable to this Engine
TBD - To Be Determined

PRECISION INDUSTRIES®
LIGHTING THE WORLD
Dubai| Abu Dhabi| Qatar| Oman| CIS Countries
Headquarter: Dubai| Tel: +971 4 338 4033| Fax: +971 4 338 3997
info@pi-dubai.com| www.pi-dubai.com